



## Scheme/ Answer Key for Valuation

*Scheme of evaluation (marks in brackets) and answers of problems/key*

**APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY**

**FIRST SEMESTER B.TECH DEGREE EXAMINATION, DECEMBER 2021**

**Course Code: CYT100**

**Course Name: ENGINEERING CHEMISTRY**

**(2019-Scheme)**

Max. Marks: 100

Duration: 3 Hours

### PART A

***Answer all questions, each carries 3 marks.***

- 1 Explanation (2) diagram (1) (3)
- 2 
$$E = 1.33 - \frac{0.0591}{6} \log \frac{0.02^2}{0.3 \times 10^{-14}} = 1.358V$$
 (3)  
Equation - 1 mark, substitution - 1 mark, answer -1 mark
- 3 a,b,c,f (Marks can be given/deducted proportionally based on the number of correct/wrong answers) (3)
- 4 Explanation (2) structure (1) (3)
- 5 Thermal stability PVC<PMMA<LDPE<PTFE -2 explanation -1 (3)
- 6 Principle -2 Any one applications -1 (3)
- 7 i)S ii) 2R,3R (1 mark each for correct answer) (3)
- 8 Definition (2) dopants (1) (3)
- 9 Total Hardness=  $200 \times 100 / 40 + 180 \times 100 / 24 = 1250$  ppm  
Temp Hardness =  $360 \times 100 / (2 \times 61) = 295$  ppm  
Perm. Hardness =  $1250 - 295 = 955$  ppm (3)  
1 mark each for total, temp. and perm. hardness
- 10 Any three differences -3 marks (3)

### PART B

***Answer one full question from each module, each question carries 14 marks***

#### Module-I

- 11 a) Glass electrode (3) Coupling with reference electrode and Nernst equation (8)  
(3) advantage (1) limitations (1)
- b) Principle (2) equations (2) applications (2) (6)
- 12 a) Principle (2) Equations (2) Procedure (2) Graph (2) (8)  
(Procedure and equation using  $Ce^{4+}$  can also be given full credits.)

$$\text{b) } 1.5175 = E^0_{\text{cell}} - \frac{2.303RT}{nF} \log Q \quad (6)$$

$$1.5175 = A + B \times 293 \text{ ----(1)}$$

$$1.5213=A+B \times 308 \text{ ----(2)}$$

## On solving

$$B = 2.5333 \times 10^{-4}$$

$$A = 1.4432 \text{ V} = E^0_{\text{cell}}$$

$$B = -\frac{2.303R}{nF} \log Q$$

$$2.5333 \times 10^{-4} = -\frac{2.303 \times 8.314}{2 \times 96500} \log Q$$

$$\log Q = -2.5535$$

$$Q = 2.795 \times 10^{-3}$$

Nernst equation -1 mark

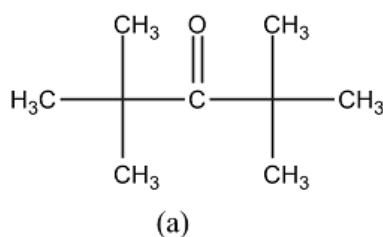
Two equation at two different temperatures -1.5 marks each

$E^0_{\text{cell}}$  -1 mark, Reaction quotient - 1 mark.

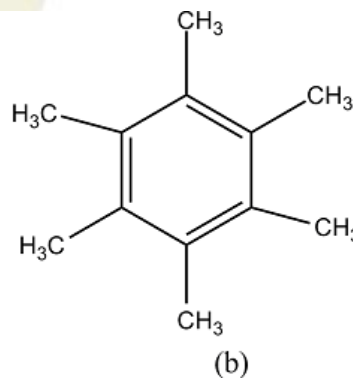
## Module-II

- 13 a) Types of transitions (4) instrumentation (4) (8)

- b) a)  $\text{DBE} = (20-18)/2=1$  b)  $\text{DBE} = (26-18)/2=4$  (6)



a) 4 marks



b) 2 marks

(Any positive attempt can be awarded with one mark each)

- 14 a) Modes-(2) sketch(4) IR activity (2) (8)

- b) State Beer-Lambert's law.(2) (6)

Define molar extinction co-efficient(1)

$A = \epsilon c x$  (Equation- 1 mark)

$$\frac{A_1}{A_2} = \frac{C_1}{C_2}$$



$$\frac{0.358}{0.518} = \frac{2.5 \times 10^{-6}}{c_2}$$

$$c_2 = 3.62 \times 10^{-6} \text{ M}$$

(2 marks)

**Module-III**

- 15 a) Instrumentation (3) Principle (1) DTA  $\text{CaC}_2\text{O}_4 \cdot \text{H}_2\text{O}$  (2 ) Explanation (2) (8)  
 b) Principle (2) block diagram (2) Instrumentation (2) (6)
- 16 a) Principle (2) Procedure(5) Solvent polarity (2) TLC monitoring (1) (10)  
 b) 4 –marks (0D,1D,2D and3D) (4)

**Module-IV**

- 17 a) 8 isomers including optically active enantiomer pair-1 mark each (8)  
 b) Definition (1) Construction (3) working (2) (6)
- 18 a) Cis and trans (3) Optical activity (1) Conformers (3) stability (1) (8)  
 b) Synthesis (3) Properties (1) applications (2). (6)  
 (3 marks can be awarded for any one methods of synthesis.)

**Module-V**

- 19 a) Principle (3) Procedure (4) calculations (3) (10)  
 b) Procedure (4) (4)
- 20 a) Action (2) application method (2) Break point chlorination graph (2) (10)  
 explanation (2), advantages (2)
- b)  $\text{CH}_2\text{O} + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$  (4)  
 30g      32g

$$\text{Wt. of oxygen} = \frac{32 \times 90}{30} = 96 \text{ mg / L } \text{ or } \text{ ppm}$$

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